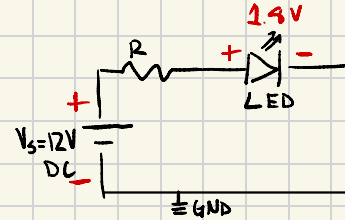


Pre-Lab Questions

- Given the manufacturers sheet for a red LED located in Canvas, **what is the forward voltage and forward current (i.e., what is the LED's operating point)?**
 - If a rating is given as a range, use the average value or midrange value.
 - You may check the *Comments* slides pdf in Canvas to verify the values needed.
- Assume a single red LED is connected in series with a resistor and connected to a $V_S = 12V$ DC power source. Use KVL and the information about the LED's current-voltage ratings to calculate **the value of R needed to operate the LED at its operating point?**
- Since there most likely won't be a resistance value sold in the particular values you will need to use the next biggest resistance. **What is the next biggest resistance value available in the following resistor kit?** Use the following link to reference available resistance values [<https://www.sparkfun.com/products/10969>].
- Next, assuming that the same voltage is across the resistor, **recalculate the current I_{LED} given the new resistance value obtained.**
- For the chosen resistance (taken from the list provided in the link) and the current that was recalculated, **compute the power through the resistor via $P = I_{LED}^2 R$.**
- For the chosen resistance (taken from the list provided in the link) and calculated power, **what is minimal power rating of the resistor that should be used (1/4, 1/2, or 1)?**

- 1.) Forward Voltage: min = 1.8V max = 2.2V
 Forward Current: Absolute Maximum Rating = 20mA
 LED's operating point: 1.8V @ 20mA

2)



$$V = IR \quad R = \frac{V}{I}$$

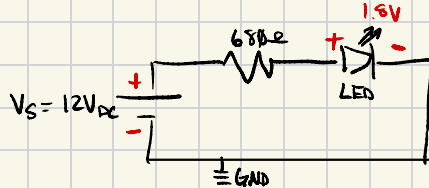
Minimum operating Point Current @ 20mA

$$R = \frac{V}{I} = \frac{12V - 1.8V}{20mA} = 510 \Omega$$

3) next Biggest resistance value = 680 Ω

4) Assuming $R = 680 \Omega$ what is I_{LED} ?

$$V = I \cdot R \rightarrow I = \frac{V}{R}$$



$$I_{LED} = \frac{V_S - V_{LED}}{R} = \frac{12V - 1.8V}{680} = 15mA$$

5) Compute the power through the resistor via $P = I_{LED}^2 R$

$$I_{LED} = 15 \text{ mA} \quad R = 680 \Omega \quad P = (15 \text{ mA})^2 (680 \Omega) = 0.153 \text{ W} = \underline{153 \text{ mW}}$$

$$6) P = 0.153 \text{ W} \quad \frac{1}{4} \text{ W} = 0.25 \text{ W}, \frac{1}{2} = 0.50 \text{ W} \quad 1 = 1 \text{ W}$$

$\frac{1}{4}$ is the minimal power rating